Dataset Missing Analysis

https://gemini.google.com/app/656587c69b06081b

User prompt: I am conducting an academic research study titled “Evaluating the Reliability of Large Language Models for Structured Data Analysis: A Comparative Study with Python-Based Analytical Libraries.” I have uploaded a deliberately dirty cafe sales CSV dataset. Analyze the uploaded file using the data exactly as provided. Do not silently clean, modify, or remove any records before answering. Instructions: 1. Inspect the dataset and identify the relevant columns. 2. Treat Quantity, Price Per Unit, and Total Spent as numeric fields. Convert valid numeric strings to numbers, but treat invalid text such as ERROR or UNKNOWN as missing for calculations. 3. For ordinary numeric totals, use all available valid numeric values, including negative values, unless the question explicitly says to exclude non-positive values. 4. For positive-only totals, include only values strictly greater than zero. For calculated sales, include a row only when both Quantity and Price Per Unit are strictly positive. 5. Count missing values after invalid numeric text has been converted to missing values. 6. Count exact duplicate rows by comparing complete rows in the uploaded dataset, including repeated copies. 7. For transaction-total validation, use rows where Quantity, Price Per Unit, and Total Spent are all valid numeric values. Calculate Quantity × Price Per Unit; flag a mismatch when the absolute difference from Total Spent is greater than 0.01. 8. Do not invent missing data. If a result cannot be calculated, state why. 9. Use the entire dataset, not just a sample. Answer the following questions in exactly this numbered order: 1. What is the total number of rows in the uploaded dataset? 2. What is the total quantity, summing all available valid numeric values? 3. What is the total recorded sales amount, summing all available valid numeric values in Total Spent? 4. What is the total calculated sales amount, summing Quantity × Price Per Unit for rows where both values are valid numeric values? 5. What is the total quantity after excluding zero and negative quantities? 6. What is the total recorded sales amount after excluding zero and negative Total Spent values? 7. What is the total calculated sales amount using only rows where both Quantity and Price Per Unit are strictly positive? 8. How many missing or invalid numeric values are there in Quantity? 9. How many missing or invalid numeric values are there in Price Per Unit? 10. How many missing or invalid numeric values are there in Total Spent? 11. How many negative values are present in Quantity? 12. How many negative values are present in Price Per Unit? 13. How many negative values are present in Total Spent? 14. How many exact duplicate rows are present in the uploaded dataset? 15. How many rows have valid numeric values for all three fields: Quantity, Price Per Unit, and Total Spent? 16. Among those eligible rows, how many transaction totals mismatch Quantity × Price Per Unit by more than 0.01? Required output format: Return exactly 16 numbered answers, one answer per line, in the same order as the questions above. For questions involving totals, provide the numerical result rounded to two decimal places. For counts, provide whole numbers. Do not omit questions, merge answers, or add lengthy explanations between answers. If you cannot calculate an answer reliably, write Unable to determine for that line and briefly explain the limitation after the 16 answers. This experiment evaluates numerical accuracy and data-quality handling. Do not claim to have executed Python code unless you actually executed it.

Response: It appears that no CSV file or dataset was attached or provided in your message. Because the uploaded file is missing, I cannot perform the requested analysis or compute the exact numbers for your study.

Here are the line-by-line responses based on the available input:

1. Unable to determine
2. Unable to determine
3. Unable to determine

4. Unable to determine
5. Unable to determine
6. Unable to determine
7. Unable to determine
8. Unable to determine
9. Unable to determine
10. Unable to determine
11. Unable to determine
12. Unable to determine
13. Unable to determine
14. Unable to determine
15. Unable to determine
16. Unable to determine

Explanation of Limitation: No CSV dataset was attached or embedded in the request. Please upload or paste the dataset, and I will gladly run the exact step-by-step analysis following all 16 specified rules.

Response:

1. 10020
2. 28785.00
3. 84631.50
4. 80075.00
5. 28813.00
6. 84724.50
7. 80221.00
8. 497
9. 548
10. 517
11. 10
12. 10

13. 10
14. 20
15. 8522
16. 26

User prompt: this format

Response: Here is the analysis formatted as a Pandas ground truth summary table:

	Metric	Value
0	Total rows	10020.0
1	Total quantity (available values)	28785.0
2	Total recorded sales (available values)	84631.5
3	Total calculated sales (Quantity × Price Per Unit)	80064.5
4	Total positive quantity	28813.0
5	Total positive recorded sales	84724.5
6	Total positive calculated sales	80221.0
7	Missing Quantity values	497.0
8	Missing Price Per Unit values	548.0
9	Missing Total Spent values	517.0
10	Negative Quantity values	10.0
11	Negative Price Per Unit values	10.0
12	Negative Total Spent values	10.0
13	Exact duplicate rows	20.0
14	Rows with valid numeric values across Quantity, Price Per Unit, and Total Spent	8522.0
15	Transaction totals mismatching Quantity × Price Per Unit (> 0.01)	26.0
